



Make Your **Home Smart** with **FOSS** and **IoT Devices**

Home sweet home – there's nothing like it! Our homes can be made smarter using the Internet of Things (IoT) – a system in which computing devices are connected over a network using unique identifiers (UIDs), and which can transfer data without the need for any human-to-human or human-to-computer interactions. These IoT devices can be used to automate our homes also.

It was back in 1975 that a network technology called X10 was developed and used in home automation for the first time. The X10 used electric power transmission wiring for signalling and controlling various electronic devices. The signals used radio frequency as digital data to control any digital electronic device installed in a house. But now, with the help of sophisticated hardware and free open source software (FOSS), home automation can make a dream house a reality.

The basic setup

The setup for home automation using IoT devices requires a router connected to the Internet and a controller/hub

that is connected to the IoT devices/sensors. There are four primary operating standards for home automation technology – Wi-Fi, Z-Wave, Zigbee, and Bluetooth Low Energy (BLE). Wi-Fi is the most commonly used standard for home automation and IoT devices are connected to a centralised controller/hub (either wired or wireless), which is connected to the router through an Ethernet link. The IoT devices can connect to a cloud network on the Internet using the controller/hub and the router. The devices can be controlled through the cloud network using a FOSS or a mobile app (Figure 1).

The mobile app can be used to send and receive data through the cloud network to control and monitor the IoT devices installed in a house.

This architecture offers users the convenience of controlling and automating various IoT devices from a distant location with the help of a smartphone and the Internet. The IoT devices are able to send data about room temperature, supply voltage, energy consumption, etc, and even provide a live video stream to the smartphone app through the Internet so that the home can be monitored remotely.

IoT devices/sensors

There are various IoT devices that can be used for home automation. Listed below are a few basic IoT devices required for home automation and security.

- **Wall switches:** IoT based wall switches can be mounted on the wall of a house and be controlled

